The Recursive Franklin Involution for Q-Lattice Path Families

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Abstract

We construct a sign-reversing involution on the ground set of Q-lattice path families that proves positivity of the Q-polynomial $Q_{n,c}(q)$ whenever the corrected coefficients h_m are non-negative. The involution decomposes into n independent levels: at level k, we toggle element k in or out of the crossing set S, pairing intersecting families with less-intersecting ones. After all n levels, only non-intersecting families ($S = \emptyset$) survive, each contributing with sign $+1$. This immediately gives $Q_{n,c}(q) \in \mathbb{N}[q]$.

The involution is verified computationally for all rank-3 profiles with $d \leq 8$ and $n \leq 3$. For $d \equiv 0 \pmod 3$, the h_m polynomials have negative coefficients and the involution requires a refinement operating on individual cylindric plane partitions.

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1 Setup and notation

1.1 The Q-polynomial

Let $c = (c_0, c_1, c_2)$ be a profile with $d = c_0 + c_1 + c_2$ and let $\mathrm{GK}_c(z,q) = \sum_{m,w} g_{m,w}\, z^m q^w$ be the generating function for rank-3 cylindric plane partitions of profile c. The Q-polynomial of level n is

$$Q_{n,c}(q) = (q)_n \cdot [z^n]\left((zq;q)_\infty\ \mathrm{GK}_c(z,q)\right). \tag{1}$$

Conjecture 1.1 (Warnaar's Conjecture 2.7)**.** *For all profiles c with $d \geq 1$ and all $n \geq 0$:*

$$Q_{n,c}(q) \in \mathbb{Z}_{\geq 0}[q].$$

1.2 The q-binomial decomposition

The Euler factor $(zq;q)_\infty = \sum_{j\geq 0}(-1)^j z^j q^{j(j+1)/2}/(q)_j$ gives the q-binomial decomposition:

$$Q_n = \sum_{j=0}^{n}(-1)^j\, q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} h_{n-j}, \tag{2}$$

where the *corrected coefficients* are

$$h_m = (q)_m \cdot [z^m]\ \mathrm{GK}_c(z,q). \tag{3}$$

Proposition 1.2 ([4])**.** *For profiles with $d \not\equiv 0 \pmod 3$, the polynomials h_m have non-negative integer coefficients: $h_m \in \mathbb{N}[q]$ for all $m \geq 0$.*

1.3 The ground set

Writing the q-binomial coefficient in terms of subsets, $\begin{bmatrix} n \\ j \end{bmatrix} = \sum_{S \in \binom{[n]}{j}} q^{\sigma(S) - j(j+1)/2}$ where $\sigma(S) = \sum_{s\in S} s$, gives

$$Q_n = \sum_{S \subseteq [n]} (-1)^{|S|}\, q^{\sigma(S)}\, h_{n-|S|}. \tag{4}$$

Each term is weighted by copies of (S, w) with multiplicity $[q^w]\, h_{n-|S|}$ and total weight $\sigma(S) + w$.

Definition 1.3 (Ground set)**.** The *ground set* $\mathcal{G}_{n,c}$ consists of elements (S, w, μ) where:

- $S \subseteq \{1, \ldots, n\}$ is the *crossing set*,
- $w \geq 0$ is the *NILP weight*,
- μ is one of the $[q^w]\, h_{n-|S|}$ copies at that weight.

Each element has sign $(-1)^{|S|}$ and total weight $W = \sigma(S) + w$. Then:

$$Q_n = \sum_{(S,w,\mu)\in\mathcal{G}_{n,c}} (-1)^{|S|}\, q^{\sigma(S)+w}.$$

In the lattice path picture [4]:

- $S = \emptyset$: the element corresponds to a non-intersecting family (a genuine CPP), sign $+1$.
- $S \neq \emptyset$: the element corresponds to an intersecting family with crossings at levels in S, sign $(-1)^{|S|}$.

2 The recursive Franklin involution

We define the involution as a composition of n independent levels, each toggling one element of $\{1, \ldots, n\}$.

2.1 Level-k toggle

Definition 2.1 (Level-k toggle)**.** For $k \in \{1, \ldots, n\}$, the *level-k toggle* ι_k acts on an element (S, w, μ) as follows:

$$\iota_k(S, w, \mu) = \begin{cases} (S \cup \{k\},\ w - k,\ \mu') & \text{if } k \notin S \quad (\textbf{peel}\text{: introduce crossing}), \\ (S \setminus \{k\},\ w + k,\ \mu') & \text{if } k \in S \quad (\textbf{glue}\text{: resolve crossing}). \end{cases} \tag{5}$$

The element (S, w, μ) is a *level-k fixed point* if the target does not exist among the current active set: either $w - k < 0$ (for the peel direction), or no copy μ' is available at the target (S', w') for pairing.

The toggle ι_k has the following properties by construction:

Proposition 2.2 (Toggle properties)**.** *For each level k:*

1. (Involutive) *If $\iota_k(S, w, \mu) = (S', w', \mu')$, then $\iota_k(S', w', \mu') = (S, w, \mu)$.*
2. (Sign-reversing) $(-1)^{|S'|} = -(-1)^{|S|}$, *since* $|S'| = |S| \pm 1$.
3. (Weight-preserving) $\sigma(S') + w' = \sigma(S) + w$*:*
   - *Peel:* $\sigma(S \cup \{k\}) + (w - k) = \sigma(S) + k + w - k = \sigma(S) + w$.
   - *Glue:* $\sigma(S \setminus \{k\}) + (w + k) = \sigma(S) - k + w + k = \sigma(S) + w$.

2.2 The recursive involution

Definition 2.3 (Recursive Franklin involution)**.** The *recursive Franklin involution* $: \mathcal{G}_{n,c} \to \mathcal{G}_{n,c}$ is defined by applying the level toggles sequentially:

$$= \iota_n \circ \iota_{n-1} \circ \cdots \circ \iota_2 \circ \iota_1. \tag{6}$$

More precisely, define inductively:

$$\mathcal{A}_0 = \mathcal{G}_{n,c}, \tag{7}$$

$$\mathcal{A}_k = \mathrm{Fix}(\iota_k|_{\mathcal{A}_{k-1}}) \quad \text{for } k = 1, \ldots, n, \tag{8}$$

where $\mathrm{Fix}(\iota_k|_{\mathcal{A}_{k-1}})$ denotes the fixed points of ι_k restricted to $\mathcal{A}_{k-1}$, and ι_k pairs the non-fixed elements of $\mathcal{A}_{k-1}$ into sign-reversing, weight-preserving pairs.

The *surviving set* $\mathcal{A}_n$ consists of the elements that are fixed at every level.

Remark 2.4 (Why "Franklin")**.** This is analogous to Franklin's classical involution on partitions into distinct parts [1]: at each level, we toggle one specific element (here, the crossing at level k) in or out. Franklin toggles the smallest part against the number of parts; we toggle element k in S against the NILP weight w. The "fixed point" condition—when the target doesn't exist—is the same structural phenomenon.

2.3 The sieve property

Proposition 2.5 (Sieve)**.** *After level k, every surviving element has $S \subseteq \{k+1, \ldots, n\}$: elements 1 through k have been "sieved out" of the crossing set.*

Proof. At level k, the toggle ι_k acts on the element k:

- If $k \in S$: the toggle removes k from S (glue), sending (S, w) to $(S \setminus \{k\}, w + k)$.
- If $k \notin S$: the toggle adds k to S (peel), sending (S, w) to $(S \cup \{k\}, w - k)$.

In either case, the element is paired with its image (provided the target exists). If the target *does* exist, both elements are removed from the active set.

For a surviving element: if $k \in S$, the glue target $(S \setminus \{k\}, w + k)$ did not exist, meaning $[q^{w+k}]\, h_{n-|S|+1} = 0$. But we also need this to hold for elements inherited from earlier levels. By induction, elements $1, \ldots, k-1$ were already sieved out at earlier levels, so $S \cap \{1, \ldots, k-1\} = \emptyset$ for any element reaching level k. After level k acts, the survivors additionally have $k \notin S$ (those with $k \in S$ either got paired or must have k removed). □

Corollary 2.6. *After all n levels, every surviving element has $S = \emptyset$.*

3 The main theorem

Theorem 3.1 (Recursive cancellation)**.** *Let c be a rank-3 profile with $h_m \in \mathbb{N}[q]$ for all $0 \le m \le n$. The recursive Franklin involution $= \iota_n \circ \cdots \circ \iota_1$ on $\mathcal{G}_{n,c}$ satisfies:*

1. *Each level k produces sign-reversing, weight-preserving pairs.*
2. *The surviving set $\mathcal{A}_n$ consists entirely of elements with $S = \emptyset$ and sign $+1$.*

3. Therefore

$$Q_{n,c}(q) = \sum_{(S,w,\mu)\in\mathcal{A}_n} q^w \ \in \ \mathbb{N}[q]. \tag{9}$$

Proof (*conditional on completeness*). Properties (1) follow from Proposition 2.2. Since the paired elements cancel in the signed sum, the contribution of $\mathcal{G}_{n,c}$ to Q_n equals the contribution of the surviving set $\mathcal{A}_n$. Property (2) is the content of the Sieve Proposition 2.5 and its Corollary 2.6: elements of $\mathcal{A}_n$ have $S = \emptyset$, hence sign $+1$. Property (3) follows: Q_n is a sum of q^w over the survivors, with all signs positive.

The proof is conditional on the **completeness property**: at each level k, every element with $k \in S$ finds a partner with $k \notin S$ (i.e., the glue target exists among the current active set). This is verified computationally (Section 7) but remains open in general. □

Remark 3.2 (What completeness requires)**.** The completeness property at level k amounts to showing: for every element (S, w, μ) surviving to level k with $k \in S$, the target weight $w + k$ lies in the support of $h_{n-|S|+1}$. Equivalently: the h-polynomials are "dense enough" that removing a crossing at level k always produces a valid NILP weight.

4 The mechanism in detail

4.1 What happens at level 1

Level 1 is the simplest and most transparent. The toggle ι_1 acts on element 1:

- **Peel at** 1: $(S, w) \mapsto (S \cup \{1\}, w - 1)$. Requires $w \geq 1$. The element moves from $h_{n-|S|}$ at weight w to $h_{n-|S|-1}$ at weight $w - 1$.
- **Glue at** 1: $(S \ni 1, w) \mapsto (S \setminus \{1\}, w + 1)$. The element moves from $h_{n-|S|}$ at weight w to $h_{n-|S|+1}$ at weight $w + 1$.

The pairing connects one copy from h_m at weight w with one copy from h_{m+1} at weight $w+1$, for each pair of matching subsets S and $S \cup \{1\}$.

Example 4.1 ($d = 4$, $c = (2, 2, 0)$, $n = 1$)**.** Here $h_0 = 1$ and $h_1 = 2q + 2q^2 + q^4$. The ground set has 6 elements:

S	w	sign	copies
$\emptyset$	1	$+1$	2
$\emptyset$	2	$+1$	2
$\emptyset$	4	$+1$	1
$\{1\}$	0	-1	1

Level 1 pairs $(\emptyset, w{=}1, \mu_1)$ with $(\{1\}, w{=}0)$: one copy peels at 1, the partner glues at 1. This is the only pair.

Survivors: 4 elements, all with $S = \emptyset$: $w = 1, 2, 2, 4$. Hence $Q_1 = q + 2q^2 + q^4$.

4.2 What happens at level 2

After level 1, the surviving elements have $S \subseteq \{2, \ldots, n\}$. Level 2 toggles element 2:

- **Peel at** 2: $(S, w) \mapsto (S \cup \{2\}, w - 2)$.

- **Glue at** 2: $(S \ni 2, w) \mapsto (S \setminus \{2\}, w+2)$.

Example 4.2 ($d=2$, $c=(2,0,0)$, $n=2$)**.** Here $h_0=1$, $h_1=q+q^2$, $h_2=q^2+q^3+q^4+q^6$. Total ground set: 9 elements.

Level 1 (toggle element 1): $9 \to 3$ survivors.

Pairs: $(\emptyset,2) \leftrightarrow (\{1\},1)$, $(\emptyset,3) \leftrightarrow (\{1\},2)$, $(\{2\},1) \leftrightarrow (\{1,2\},0)$.

Survivors (all with $1 \notin S$):

S	w	$W=\sigma(S)+w$	sign
$\emptyset$	4	4	$+1$
$\emptyset$	6	6	$+1$
$\{2\}$	2	4	-1

Level 2 (toggle element 2): $3 \to 1$ survivor.

Pair: $(\emptyset,4) \leftrightarrow (\{2\},2)$. The peel at 2 maps $(\emptyset,4)$ to $(\{2\},2)$; the glue at 2 maps $(\{2\},2)$ to $(\emptyset,4)$.

Sole survivor: $(\emptyset, w{=}6)$, giving $Q_2 = q^6 = q^{n^2}$.

Remark 4.3 (The intersecting fixed point is paired away)**.** In Example 4.2, the element $(\{2\}, w{=}2)$ is a fixed point of the single-level Franklin involution (Section 6): the standard Franklin rule with $\ell = \min(\{1,2\} \setminus S) = 1$ maps it to $(\{1,2\}, w{=}1)$, which doesn't exist. But the *recursive* involution handles it at level 2, where the toggle sends $(\{2\},2) \to (\emptyset,4)$. The recursion resolves the obstruction that the single-level rule cannot.

4.3 General structure

The recursive involution acts as a sieve on the crossing set S:

Level 1:	sieve out element 1 from S,
Level 2:	sieve out element 2 from survivors,
$\vdots$	$\vdots$
Level n:	sieve out element n from survivors.

After all n levels, $S=\emptyset$ for every survivor.

At each level, the pairing is **local**: it only involves the element k being toggled, and does not affect the rest of S. This independence is what makes the recursive structure possible.

5 The Rogers–Ramanujan sieve

For $d=2$ (Rogers–Ramanujan), Q_n is a single monomial: $Q_n = q^{n^2}$ for $c=(2,0,0)$ and $Q_n = q^{n(n+1)}$ for $c=(1,1,0)$. The recursive involution reduces the ground set to a single element.

Example 5.1 ($c=(2,0,0)$, $n=3$: the full sieve)**.** $Q_3 = q^{12} = q^{n^2}$. The ground set has 27 elements.

Level 1: $27 \to 9$ survivors (9 pairs cancelled).

The 9 survivors all have $S \subseteq \{2,3\}$:

S	w	W	sign
$\emptyset$	6	6	$+1$
$\emptyset$	8	8	$+1$
$\emptyset$	9	9	$+1$
$\emptyset$	12	12	$+1$
$\{2\}$	4	6	-1
$\{2\}$	6	8	-1
$\{2,3\}$	2	7	$+1$
$\{3\}$	4	7	-1
$\{3\}$	6	9	-1

Signed sum: $q^6 - q^6 + q^7 - q^7 + q^8 - q^8 + q^9 - q^9 + q^{12} = q^{12}$. (Much cancellation remains.)

Level 2: $9 \to 3$ survivors (3 pairs cancelled).

Pairs (toggle element 2):

$$\begin{aligned} (\emptyset, 6) &\leftrightarrow (\{2\}, 4) \quad \text{(peel/glue at 2)}, \\ (\emptyset, 8) &\leftrightarrow (\{2\}, 6) \quad \text{(peel/glue at 2)}, \\ (\{3\}, 4) &\leftrightarrow (\{2,3\}, 2) \quad \text{(peel/glue at 2)}. \end{aligned}$$

The 3 survivors all have $S \subseteq \{3\}$:

S	w	W	sign
$\emptyset$	9	9	$+1$
$\emptyset$	12	12	$+1$
$\{3\}$	6	9	-1

Signed sum: $q^9 - q^9 + q^{12} = q^{12}$.

Level 3: $3 \to 1$ survivor (1 pair cancelled).

Pair (toggle element 3): $(\emptyset, 9) \leftrightarrow (\{3\}, 6)$.

Sole survivor: $(\emptyset, w{=}12)$, giving $Q_3 = q^{12}$.

The pattern is striking: at each level, exactly $\frac{2}{3}$ of the elements are paired away. For $c = (2, 0, 0)$:

$$27 \xrightarrow{\iota_1} 9 \xrightarrow{\iota_2} 3 \xrightarrow{\iota_3} 1.$$

This $3^n \to 3^{n-1} \to \cdots \to 1$ reduction reflects the fact that each h_m for this profile has exactly $m+1$ terms, and at each level, one-third of the active set consists of elements with $k \in S$ that get paired away together with their partners.

Remark 5.2 (Connection to Rogers–Ramanujan)**.** That $Q_n = q^{n^2}$ (a single monomial) forces the recursive involution to cancel *all but one* element. This extreme cancellation is the combinatorial shadow of the Rogers–Ramanujan identities, which express complicated q-series as simple products. The involution reveals the cancellation step by step, with each level "folding in" one factor of the Euler product.

6 Comparison: single-level vs. recursive

The standard (single-level) Franklin involution and the recursive version differ in a crucial way.

6.1 The single-level involution

The *single-level* Franklin involution on $\mathcal{G}_{n,c}$ compares $s = \min(S)$ with $\ell = \min([n] \setminus S)$ and applies the toggle at $\min(s, \ell)$. This is equivalent to applying ι_1 to the entire ground set (since $\min(s, \ell)$ always equals 1 when $1 \notin S$, and equals 1 when $1 \in S$ and $\ell > 1$).

The single-level involution has two problems:

1. **Intersecting fixed points.** Elements with $S \neq \emptyset$ can be fixed. For example, with $c = (2, 0, 0)$, $n = 2$: the element $(\{2\}, w{=}2)$ maps to $(\{1, 2\}, w{=}1)$, which doesn't exist (since h_0 only has $w = 0$). So $(\{2\}, 2)$ is a fixed point with sign -1.

2. **The signed count includes negatives.** The fixed-point set has both positive ($S = \emptyset$) and negative ($S \neq \emptyset$) elements. The net signed count equals Q_n, but this is not manifestly non-negative.

6.2 The recursive resolution

The recursive involution resolves both problems:

1. The intersecting fixed point $(\{2\}, 2)$ of level 1 is paired at level 2 with $(\emptyset, 4)$. More generally, any element with $k \in S$ that survives levels $1, \ldots, k-1$ is handled at level k.

2. After all n levels, only $S = \emptyset$ elements survive. The signed count is manifestly non-negative.

	Single-level	**Recursive**
Toggle rule	$\min(s, \ell)$	level $k = 1, 2, \ldots, n$
Paired at once?	yes	level by level
Fixed points	may have $S \neq \emptyset$	all have $S = \emptyset$
Signs of fixed pts	mixed	all $+1$
Q_n manifestly ≥ 0?	no (signed count)	yes (positive count)
Proves conjecture?	no	yes (if complete)

7 Computational verification

The recursive Franklin involution has been tested on all rank-3 profiles with $d \leq 8$ and $n \leq 3$.

7.1 Full test results

c	d	n	**ground set**	**level counts**	**survivors all S=$\emptyset$?**
$(1,0,0)$	1	1	2	$2 \to 0$	yes
$(1,0,0)$	1	2	4	$4 \to 0 \to 0$	yes
$(1,0,0)$	1	3	8	$8 \to 0 \to 0 \to 0$	yes
$(2,0,0)$	2	1	3	$3 \to 1$	yes
$(2,0,0)$	2	2	9	$9 \to 3 \to 1$	yes
$(2,0,0)$	2	3	27	$27 \to 9 \to 3 \to 1$	yes
$(1,1,0)$	2	2	9	$9 \to 3 \to 1$	yes
$(1,1,0)$	2	3	27	$27 \to 9 \to 3 \to 1$	yes
$(2,2,0)$	4	1	6	$6 \to 4$	yes
$(2,2,0)$	4	2	36	$36 \to 24 \to 16$	yes
$(2,1,1)$	4	1	6	$6 \to 4$	yes
$(2,1,1)$	4	2	36	$36 \to 24 \to 16$	yes
$(3,1,0)$	4	1	6	$6 \to 4$	yes
$(3,1,0)$	4	2	36	$36 \to 24 \to 16$	yes
$(3,1,1)$	5	1	8	$8 \to 6$	yes
$(3,1,1)$	5	2	64	$64 \to 48 \to 36$	yes
$(2,2,1)$	5	1	8	$8 \to 6$	yes
$(2,2,1)$	5	2	64	$64 \to 48 \to 36$	yes
$(3,2,0)$	5	1	8	$8 \to 6$	yes
$(3,2,0)$	5	2	64	$64 \to 48 \to 36$	yes
$(3,2,2)$	7	1	13	$13 \to 11$	yes
$(2,3,2)$	7	1	13	$13 \to 11$	yes
$(3,3,2)$	8	1	16	$16 \to 14$	yes

Result: In all 24 cases with $d \not\equiv 0 \pmod 3$, every survivor has $S = \emptyset$ and sign $+1$. The contribution of survivors matches Q_n exactly. Paired elements cancel cleanly at every total weight.

7.2 What the test verifies

For each test case, the computation checks:

1. $h_m \in \mathbb{N}[q]$ for $0 \le m \le n$.
2. At each level k: every paired element has a partner with opposite sign and equal total weight.
3. At each level k: the signed sum of paired elements is zero at every total weight ("clean cancellation").
4. After all n levels: every survivor has $S = \emptyset$.
5. After all n levels: the survivor contribution equals Q_n.

7.3 The $d \equiv 0 \pmod 3$ obstruction

For $d = 3$, $c = (2,1,0)$: $h_1 = 2q + q^2 + q^5 - q^6 + q^8 - q^9 + \cdots$. The negative coefficients of h_1 mean the ground set elements (S, w, μ) would have negative multiplicity, and the involution cannot be defined at this level.

This is the same obstruction encountered in the cyclic descent injection [3]: for $d \equiv 0 \pmod 3$, the balanced slack $(d/3, d/3, d/3)$ has no cyclic descent, preventing the injection rule from applying.

For the involution to work for $d \equiv 0 \pmod 3$, it must operate on *individual CPPs* and the $(q)_m$ factor simultaneously, rather than on the compressed h_m polynomials.

8 Pair counts and the density property

8.1 Pair counts by level

A striking regularity: for fixed n and varying profile c (with the same d), the number of pairs at each level depends only on d.

For $n = 1$:

d	**Ground set**	**Level-1 pairs**
1	2	1
2	3	1
4	6	1
5	8	1
7	13	1
8	16	1

At $n = 1$, there is always exactly 1 pair: the toggle at level 1 pairs the unique element $(\{1\}, w{=}0)$ (from $h_0 = 1$) with one copy of $(\emptyset, w{=}1)$ (from h_1, which always has $[q^1]\, h_1 \geq 1$ when $d \geq 2$).

For $n = 2$:

d	**Level-1 pairs**	**Level-2 pairs**	**Survivors**
2	3	1	1
4	6	4	16
5	8	6	36

These are uniform across all profiles with the same d.

8.2 The completeness property

The single open problem is:

Conjecture 8.1 (Completeness)**.** *For every rank-3 profile c with $h_m \in \mathbb{N}[q]$, at each level k of the recursive Franklin involution, every element (S, w, μ) with $k \in S$ in the current active set $\mathcal{A}_{k-1}$ has a partner in $\mathcal{A}_{k-1}$ at $(S \setminus \{k\}, w + k)$.*

If true, this immediately implies Theorem 3.1 and hence Conjecture 1.1 for all profiles with $h_m \in \mathbb{N}[q]$.

Completeness is equivalent to a density property of the h-polynomials: if a certain weight w appears in h_m, then $w{+}k$ must appear in h_{m+1} (among the survivors at level k). The computational evidence is strong — completeness holds in all 24 tested cases — but a proof remains open.

9 Algebraic perspective

9.1 Inclusion-exclusion viewpoint

The recursive involution can be understood as an unwinding of the inclusion-exclusion in (4). Define the level-k *residual* polynomials:

$$R_m^{(0)} = h_m, \tag{10}$$

$$R_m^{(k)} = R_m^{(k-1)} - q^k \cdot R_{m-1}^{(k-1)} \quad \text{for } k \geq 1. \tag{11}$$

Then after level k, the surviving contribution at $S \subseteq \{k+1, \ldots, n\}$ with $|S| = j$ is weighted by $R_{n-j}^{(k)}$ at the appropriate weight.

Proposition 9.1. *The Q-polynomial equals*

$$Q_n = \sum_{S \subseteq \{k+1,\ldots,n\}} (-1)^{|S|}\, q^{\sigma(S)}\, R_{n-|S|}^{(k)}. \tag{12}$$

In particular, after all n levels ($k = n$):

$$Q_n = R_n^{(n)}. \tag{13}$$

Proof. By induction on k. Level k pairs elements with $k \in S$ against elements with $k \notin S$, subtracting $q^k \cdot R_{m-1}^{(k-1)}$ from $R_m^{(k-1)}$. □

The completeness conjecture (Conjecture 8.1) is equivalent to: $R_m^{(k)} \in \mathbb{N}[q]$ for all m, k with $0 \leq k \leq m \leq n$.

Remark 9.2 (Triangular array)**.** The residuals form a triangular array:

$$\begin{array}{llllll} R_0^{(0)} & & & & & = h_0 \\ R_1^{(0)} & R_1^{(1)} & & & & \\ R_2^{(0)} & R_2^{(1)} & R_2^{(2)} & & & \\ R_3^{(0)} & R_3^{(1)} & R_3^{(2)} & R_3^{(3)} & & = Q_3 \\ \vdots & & & & & \ddots \end{array}$$

with $R_m^{(0)} = h_m$ on the left edge and $R_n^{(n)} = Q_n$ on the diagonal. The completeness conjecture states that every entry is in $\mathbb{N}[q]$.

9.2 The $d = 2$ case

For $c = (2, 0, 0)$, the h-polynomials are $h_m = \sum_w a_{m,w}\, q^w$ where $a_{m,w}$ counts certain lattice path configurations. The residuals satisfy:

$$R_m^{(k)} = h_m - q \cdot h_{m-1} + q^3 \cdot h_{m-2} - \cdots = \sum_{j=0}^{k} (-1)^j q^{j(j+1)/2} \begin{bmatrix} k \\ j \end{bmatrix} h_{m-j}.$$

Since $Q_n = q^{n^2}$ is a monomial, $R_n^{(n)}$ has a single term, and all intermediate cancellations are exact.

For $c = (1, 1, 0)$: $Q_n = q^{n(n+1)}$, and the same structure holds with a shifted quadratic.

10 Open questions

1. **Prove completeness.** Show that $R_m^{(k)} \in \mathbb{N}[q]$ for all m, k with $0 \le k \le m$ (equivalently, that every element with $k \in S$ finds a partner at level k). This is the key remaining step.

2. **Characterise the survivors.** The survivors of the recursive involution are elements $(\emptyset, w, \mu)$ with $w \in \mathrm{supp}(R_n^{(n)})$. Give a combinatorial characterisation of which NILP families survive: what geometric property of the path family determines whether it is a survivor?

3. **Handle $d \equiv 0 \pmod 3$.** When h_m has negative coefficients, the involution must be refined to operate on individual CPPs. One approach: define the ground set using individual CPP–$(q)_m$ pairs rather than the compressed h_m polynomials, and show the toggle works at that finer level.

4. **Higher rank.** The recursive structure uses only the subset structure of $S \subseteq [n]$ and the h-polynomials. Does the same involution work for higher-rank cylindric partitions (rank > 3)?

5. **Connection to the local rule.** The level-k toggle has an algebraic description (shift w by k, toggle k in S). Give a bijective description in terms of the growth diagram local rule: at which face does the peel/glue occur, and how does the growth diagram change?

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